Burnt, Not Gone: A Photometric Emission Gate for Tracking Through Sensor Saturation

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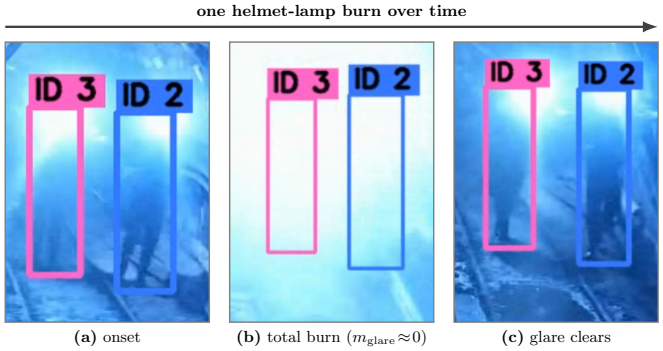
***Abstract*—A helmet headlamp pointed at a fixed mine camera saturates the sensor and erases a worker's image in place, so the body detector returns nothing and the track is dropped. The position of a burnt worker is not the hard part: a near-static worker is predicted well by a constant-velocity Kalman coast, and a coast allowed to run through the burn recovers the body on most frames. The hard part is *deciding whether to keep emitting the box at all*: present-but-burnt, or departed? A coast that runs blindly answers "always," so it recovers burnt workers only at a heavy false-positive cost. We show that this decision can be made directly from the destroying signal: the fraction of the predicted body box that is photometrically saturated. A one-threshold gate on this fraction emits the coasting box only while the body is actually burnt, recovering 77.6% of total-burn frames where every motion, appearance, learned-motion and amodal-occlusion baseline scores 0.0%, and on full-scene synthetic-burn video it keeps a blind coast's burn recall to within two points (85.61 vs. 87.4) while cutting its false positives from 4596 to 2024 (a 56% reduction), so that, despite recovering burnt workers, it attains a *higher* MOTA (73.24) than not recovering at all (71.7). The emission decision is a single analytic threshold, with no learning and no appearance model, layered on a glare-aware detection front-end, and runs online at 26.0 fps; an operating-point sweep shows it is insensitive to its threshold, and an ablation shows that more elaborate detectors (a matched-filter headlamp confirmation, a sequential CUSUM test, an innovation gate) do not improve it.**

***Index Terms*—multi-object tracking, sensor saturation, photometric gating, amodal recovery, industrial safety**

I. Introduction

Multi-object tracking (MOT) for worker localization in an underground mine runs in a hard photometric regime. Cameras are fixed, workers wear near-identical protective equipment, and helmet lamps repeatedly point into the camera and clip the sensor. While a lamp faces the camera the worker's pixels are photometrically destroyed where they stand: luminance saturates and chroma collapses, so the body detector loses the box and an appearance descriptor is uninformative. Tracking-by-detection systems [1]–[3] respond to a missing detection by coasting a constant-velocity prediction for a fixed number of frames and then terminating. In a safety setting (localization, collision avoidance, emergency response) the dropped track and fragmented identity is the worst outcome.

It is tempting to treat this as localization, re-estimating where the burnt worker is, but on this data inertia largely solves it: a burnt worker is typically near-static, so a constant-velocity Kalman coast predicts the amodal box well, recovering the target on most burn frames (Section V). The difficulty is elsewhere. A coast that is allowed to run also runs for *every* lost track, whether a worker who walked out of frame, a body briefly occluded by a pipe, or a detector flicker, and emits a box for each, so it buys burn recovery with a flood of false positives. The question a tracker must answer under saturation is therefore not *where* but *whether*: is this missing detection a recoverable burn, or has the target gone?



(a) onset (b) total burn ($m_{\text{glare}} \approx 0$) (c) glare clears

Fig. 1. **Keeping two workers through a total headlamp burn.** A helmet lamp saturates the fixed camera and destroys both workers' image evidence (b): the body detector fails. A constant-velocity coast still predicts the near-static body, but a blind coast cannot tell a burnt worker from one who has left. Our photometric emission gate (PEG) emits the coasting box only while the predicted body box is photometrically saturated, and both identities re-emerge when the glare clears (c).

We answer it with the destroying signal itself. A burn is not an absence of information; it leaves a strong, measurable mark, a washed-out clipped region over the body, exactly while the body channel is dark. For each lost track we measure the fraction of its predicted body box that is photometrically saturated and emit the coasting box only while that fraction is high. The box keeps riding the Kalman coast; the gate only decides whether it is shown. This one-threshold analytic rule turns out to recover burnt workers at almost no cost in identity and association accuracy, and we find that more elaborate detectors layered on top of it (a matched-filter headlamp confirmation, a sequential change test, an innovation gate) do

not help. Our contributions are:

- we show that persistence-through-saturation is a *detection* problem (burnt vs. departed), and that the limiting cost of recovery is *false positives*, not box recall;
- the **photometric emission gate (PEG)** and its evidence-gated coast: amodal emission is decided by one analytic photometric quantity, the saturated fraction of the predicted body box, with a single threshold and no learning in the decision itself, adding no appearance model on top of the detector; and
- an evaluation in false-positive/false-negative terms, with an operating-point sweep and an ablation, showing that the gate recovers burnt workers at 56% fewer false positives than a blind coast, at a *higher* MOTA than not recovering at all, and that elaborate detectors add nothing.

II. Related Work

Online MOT. Tracking-by-detection pairs a detector with a motion filter and an association step: SORT [1], ByteTrack [2], OC-SORT [3], and Deep-OC-SORT, BoT-SORT, Hybrid-SORT and TrackTrack which add appearance, camera-motion compensation or track-aware association [4]–[7]; DiffMOT replaces the motion prior with a learned diffusion predictor [8]. All decide track termination by a fixed maximum age and ignore the image evidence of *why* a detection is missing.

Tracking through missing observations. Amodal trackers forecast a body hidden by a foreground occluder from motion and depth ordering [9]. They assume the evidence is displaced by an occluder, which is false under saturation: there is no occluder, no low-score box, and the worker's pixels are gone at the sensor.

Saturation and glare in imaging. HDR reconstruction [10] and lens-flare removal [11] treat sensor clipping as something to *undo*; we take the opposite stance and *read* the clipped region as positive evidence that a present worker is burnt, so no pixel recovery is needed.

Detection and estimation. The elaborate variants we test are classical: the matched filter is the optimal detector for a known signal in noise [12]; the cumulative-sum test is the standard sequential change detector [13]; innovation gating with a chi-square test is the textbook way to reject spurious measurements [14]. Our finding is that, for this gating decision, none improves on a single photometric threshold. We evaluate with HOTA, CLEAR-MOT and identity metrics under MOTChallenge conventions [15]–[17].

III. Data

Studying tracking under headlamp saturation requires footage in which the phenomenon actually occurs, and to our knowledge no public MOT benchmark offers fixed-camera footage with frequent whole-body lamp burns. We therefore assembled our own working set from the recordings of an operational site's existing occupational-safety cameras (≈22 fps, 1536×1536), annotated in the MOTChallenge nine-column format: a full-body box and a stable identity are kept whenever a worker is physically present (including saturated frames) by interpolating from the clearly visible frames that bracket each burn, which supplies the amodal ground truth a burn would otherwise destroy. **Synthetic burns with known severity.** A real burn gives no ground truth for *how* saturated a worker is at each instant, which a recovery study needs. Over held-out, detector-strong sequences we composite 1–3 non-overlapping burn events, each on a present worker, with a smooth raised-cosine attack/hold/decay envelope (≤ 3 s) defining a per-frame severity $m(t) \in [0, 1]$. The lamp is rendered in linear light (a white-hot core, an $m(t)$-scaled blue-white bloom, faint spikes and a ghost) to match a real mine lamp, not a generic starburst. Only pixels change (the amodal box and identity are kept and the schedule logged), so each burnt sequence is a pixel-aligned twin with known severity. Fig. 2 shows one event: as $m(t)$ ramps, the measured ϕ_t of the worker's box rises to 1 and the detector loses the body, yet ϕ_t stays above the gate threshold. The footage was recorded under the site operator's authorization, with workers notified of monitoring and identities reduced to sequence-local integers that carry no biometric content; in keeping with these privacy and governance obligations the material can be made available to researchers on request. We treat it here purely as the experimental substrate for the tracking question under study rather than as a benchmark in its own right.

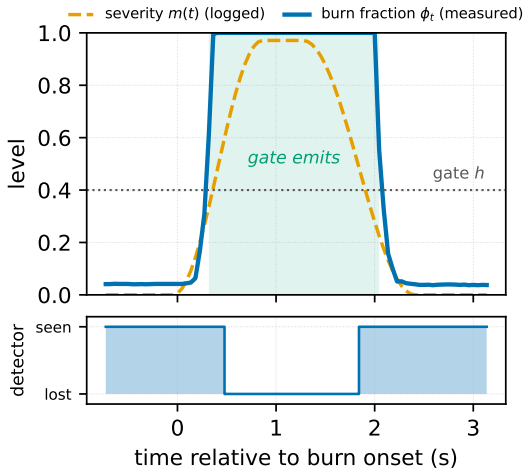


Fig. 2. **Anatomy of one synthetic burn event** (logged target, ∼2.4 s). *Top*: the generative severity envelope $m(t)$ (raised-cosine, from the dataset schedule) and the *measured* burn fraction ϕ_t of the worker's body box; the shaded band is where $\phi_t \geq h$, i.e. where the photometric gate keeps the track emitted. *Bottom*: whether the shared detector still returns the body box. As the lamp ramps up the body saturates ($\phi_t \to 1$) and the detector loses it, but ϕ_t stays high, the analytic evidence that lets PEG hold the identity through the burn.

IV. Method

PEG has three parts (Fig. 3): a glare-aware perception front-end that turns each frame into boxes annotated with a photometric reliability, an OC-SORT motion backbone that associates and coasts them, and (the contribution) a photometric emission gate that decides, on a missed frame, whether to keep emitting a track's box. The tracker is appearance-free:

it never extracts a re-identification embedding. While a track is matched to a detection it behaves as the ordinary backbone; the gate acts only when a confirmed track has *no* matched detection, where a standard tracker would blindly coast for a fixed age.

A. Glare-aware perception front-end

A shared YOLO detector [18] produces the body boxes. A small *visibility head* then reads, for each box, the detector's own frozen feature map (the stride-8 P3 tap, 256 channels) roi-pooled to 7×7, passes it through two 3×3 convolutions and a two-layer MLP, and emits three quantities: a presence logit, a reliability $m \in [0,1]$, and a 2×2 localisation covariance Σ. The tracker uses only the first two; Σ is trained for an optional weak-update channel that the emission gate below does not use.

The two signals have concrete meanings. *Presence* separates a real worker from environment light (a floor reflection, a wall streak, a lamp ghost) that the detector also fires on under glare. *Reliability* m is high for an intact body and falls as it washes out; it regresses an *analytic* target, the torso-weighted unsaturated fraction of the box (weighted lightly at the head, where the lamp sits, fully over torso and legs), so the head sharpens a physical quantity rather than learning one from scratch.

Training is pool-first: with the detector frozen, the roi-pooled features are precomputed once and only the head is fit, by minimising

$$\mathcal{L} = \lambda_p \mathcal{L}_{\text{pres}} + \lambda_m \mathcal{L}_m + \lambda_\Sigma \mathcal{L}_\Sigma. \tag{1}$$

Here $\mathcal{L}_{\text{pres}}$ is an α-balanced focal cross-entropy [19] on presence (α=0.75, γ=2; positives are workers, negatives are environment-light boxes with mined hard lamp-like negatives), $\mathcal{L}_m$ a Huber term tying m to its analytic target, and $\mathcal{L}_\Sigma$ a 2-D Gaussian negative log-likelihood for Σ, weighted $(2, 1, 0.5)$ (AdamW, 100 epochs; backbone frozen throughout). At inference the tracker reads m and presence to gate detection trust, birth, confirmation and re-entry, so glare artefacts do not spawn identities.

Two consequences matter. First, this front-end is *learned*, so PEG as a whole is *not* a learning-free system; only the emission decision (Section IV-B) is. Second, the head reads body features, so during a *total* burn, when those features are destroyed, it is uninformative; the recovery of fully burnt workers (Section V) therefore rests on the analytic gate, not the learned head.

B. The photometric emission gate

At frame t let $V_t(p), S_t(p)$ be the HSV value and saturation at pixel p. The saturated set

$$\Omega_t = \{p : V_t(p) \geq \tau_V,\ S_t(p) \leq \tau_S\} \tag{2}$$

collects the near-clipped, low-chroma pixels of a clipping light source. For a lost track with predicted body box b_t we form the *burn fraction*

$$\phi_t = \frac{|\Omega_t \cap b_t|}{|b_t|} \in [0,1], \tag{3}$$

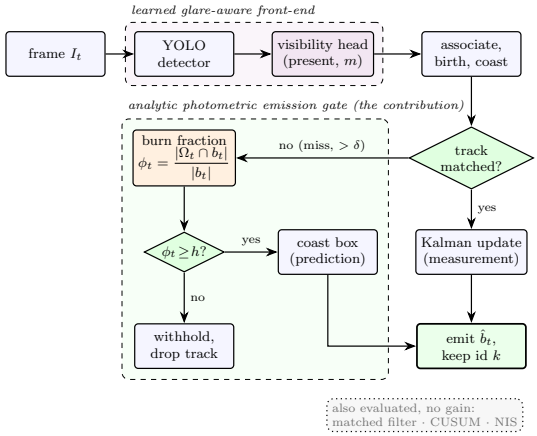


Fig. 3. **The PEG pipeline.** A shared detector and a learned visibility head (presence, reliability m) feed the OC-SORT association and birth/coast logic; this front-end is learned. When a confirmed track is matched, it is the ordinary Kalman tracker. When it is *not*, the contribution acts: a brief miss is bridged for δ frames, after which the analytic photometric emission gate emits the coasting box iff the predicted body box is saturated ($\phi_t \geq h$) and otherwise withholds it. The dashed block (matched-filter confirmation, CUSUM, NIS) is evaluated but not needed. No re-identification embedding is used anywhere.

the share of the box that is photometrically destroyed. A burnt worker drives ϕ_t high exactly while the detector fails; a worker who left, or one occluded by a non-emitting structure, leaves ϕ_t low. The gate makes the burn fraction the arbiter of emission. A very brief miss (up to a short floor of δ frames, absorbing a single detector flicker or association hiccup) is bridged as in any tracker; once a track has been unmatched longer than that, its coasting box is emitted iff

$$\phi_t \geq h, \tag{4}$$

re-evaluated every frame, with a short grace window so a one-frame dip does not drop a genuine burn. This is the decisive difference from a fixed-age coast, which emits every lost track up to its cap regardless of evidence: here the destroying signal itself decides which coasts are worth holding. The emitted box is the Kalman coast, which already tracks the near-static body, and the identity is preserved, so the worker re-emerges under the original id. The gate adds one threshold, a short floor, and a per-box sum over a binary map.

C. Refinements that do not help

Three classical detectors can be layered on the gate (each evaluated in Section V): (i) a *matched-filter* confirmation requiring a compact headlamp core in the predicted head band (the optimal correlation detector for a Gaussian-blurred point source [12]); (ii) a *CUSUM* sequential test [13] replacing the instantaneous threshold (4) by $g_t = \max(0, g_{t-1} + \phi_t - \nu)$, $g_t > h$; and (iii) an *NIS* gate [14] checking the lamp-implied body centre against the Kalman prediction. None improves the simple gate (Table IV): the synthetic burns saturate the whole body, so ϕ_t alone is already decisive, while the sequential test adds latency that costs false positives.

Algorithm 1 Photometric emission gate (per missed track).

Require: confirmed track k, gap g since last match, predicted box b_t, sat. map Ω_t

if $g \le \delta$ **then emit** Kalman coast box, keep id k ▷ brief-miss bridge
else
  $\phi_t \leftarrow |\Omega_t \cap b_t| / |b_t|$ ▷ burn fraction
  if $\phi_t \ge h$ **then emit** coast box, keep id k ▷ burnt: recover
  else withhold; drop once outside the grace window ▷ departed
  end if
end if

D. Parameters

Thresholds are fixed on validation and held across the test sequences: saturated-pixel $(\tau_V, \tau_S) = (0.92, 0.35)$, gate $h = 0.40$, brief-miss floor $\delta = 3$ frames, grace window 5 frames. Reusing the detector forward pass, the pipeline runs online at 26.0 fps on a single RTX 5070 Ti.

V. Experiments

Protocol. All methods are online and share the same YOLO11m detector [18] at 1536 px (confidence 0.25), the same scoring code and the same sequences; appearance baselines use the same fine-tuned FastReID [20], which on this identical PPE is near-useless (mean cosine 0.985 within *and* between identities, margin 0.000, 1,404 crops): the appearance cost is then near-constant across candidates and cannot change an assignment, so PEG uses no re-ID. PEG also runs the learned visibility head of Section IV-A on top of that shared detector, which the other trackers do not; we therefore read the real-test comparison (Table I) as "detector + glare-aware front-end + gate" versus "detector + tracker," and isolate the gate's own effect in the false-positive and recovery experiments below, where the head is uninformative because the body is burnt out. We report HOTA with DetA/AssA [15], IDF1 [17] and CLEAR-MOT [16] via TrackEval. DiffMOT uses the authors' checkpoint; DIP is a best-effort reproduction fed the same detections, re-ID and a depth prior [9], [21].

Real held-out test (Table I). Every method shares one frozen detector, so the spreads are modest, but PEG leads on HOTA, DetA, AssA and IDF1 and has the fewest identity switches (17 versus 27–113, a 1.6–6.6× reduction), with no appearance model. It does not lead on MOTA (PEG 80.62 versus 80.8–81.8 for the four strongest baselines): the evidence-gated coast is deliberately more permissive than emitting nothing once a track is lost, so it carries a few more false positives: holding the box lifts recall and identity (HOTA, IDF1) while MOTA counts the extra boxes directly, and the real test holds little total burn, so this residual cost is small.

Recovery is a false-positive problem (Table II). On the synthetic-burn twins (15 held-out sequences, 30 burn events) we measure full-scene MOT, so a coasting box on a departed worker counts as a false positive. Recovery recall is *generic*: a blind hold that lets OC-SORT or ByteTrack emit the coasted box through the burn recovers it about as well as PEG, since a near-static body is predicted by any coast. Its cost is not: those ungated boxes flood false positives (2.5–5× PEG's) and collapse MOTA. PEG instead keeps the recall by emitting *only* while the body is saturated, at 2024 false positives (a 56% reduction over a blind cap-90 coast) and the *best* MOTA (73.24), above even a coast that recovers nothing (71.7).

TABLE I
Real held-out test MOT (20 sequences). Same YOLO11m detector at confidence 0.25; Re-ID rows use the same fine-tuned FastReID. PEG uses no re-identification but does add the learned visibility head of Section IV-A on top of the shared detector (see Protocol). Best per column in bold.

Method	HOTA ↑	DetA ↑	AssA ↑	IDF1 ↑	MOTA ↑	IDSW ↓	Re-ID
PEG (ours)	**68.20**	**65.89**	**71.62**	**82.30**	80.62	**17**	✗
ByteTrack [2]	66.22	64.40	68.99	79.70	80.26	27	✗
OC-SORT [3]	65.46	65.51	66.64	76.44	80.82	79	✗
Deep-OC-SORT [4]	64.68	65.54	65.02	75.08	80.86	83	✓
DiffMOT [8]	66.64	65.27	69.02	79.75	81.08	51	✓
BoT-SORT [5]	66.59	64.63	69.58	79.69	80.30	29	✓
HybridSORT [6]	66.40	65.02	68.86	78.62	80.32	64	✓
TrackTrack [7]	67.36	65.61	70.17	81.14	**81.79**	29	✓
DIP [9]	62.31	62.51	63.23	72.80	75.70	113	✓

TABLE II
Recovery recall is generic; its cost is not. A blind hold lets two strong baselines emit the coasted box through the burn and recover it about as well as PEG, but their ungated boxes flood false positives and collapse MOTA; the gate keeps the recall at a fraction of the cost. Rcv@.5 on burnt targets, FP/MOTA full-scene; best per column in bold.

Method	Rcv@.5 ↑	FP ↓	MOTA ↑
OC-SORT, blind coast	73.1	10,289	26.7
ByteTrack, blind coast	73.5	5,121	55.9
PEG (gated)	**76.5**	**2024**	**73.24**

Recovery through burn (Table III). Scoring only the burnt target during its burn, stratified by remaining visibility, every baseline scores 0.0% under total burn *at its default emission policy* (DIP reaches 1.4% under heavy burn); this zero is an emission-policy artifact, not a capability gap, as the blind-coast baselines of Table II confirm. PEG recovers 74.1%/77.6% of heavy/total-burn frames by emitting *only* where the saturated fraction confirms a burn, usually under the entering identity (ID-off 18.5% versus 93–97%); a clear-reference control ($m \ge 0.9$) confirms these targets are detectable when unburnt, so the gap is caused by the burn.

Operating point and ablation (Table IV). Sweeping h from 0.2 to 0.6 leaves burn recall flat (85.6–85.8%) while MOTA rises 72.0 → 73.8 as false positives fall: a burn saturates the whole body, so ϕ_t is decisive and the gate is insensitive to h (we fix h=0.4 on validation). The classical refinements do not pay: the matched filter is neutral, while CUSUM and the full stack add false positives (the sequential lag holds boxes past the burn). A leave-one-out from the full stack agrees: only removing CUSUM helps (71.6 → 73.4), at best matching the bare gate (73.2). Removing the gate is the capped coast: no recovery (recall 83.2 versus 85.6) at a lower MOTA.

Stress test: departure into glare. To probe burnt-vs-departed on the hard negatives the standard twins lack, we composite over *empty* boxes either a broad low-chroma reflection or a foreign headlamp (compact core) and test against

TABLE III
RECOVERY ON BURNT TARGETS ONLY (SYNTHETIC GLARE TWINS). RCV IS RECOVERY@IOU0.5 AGAINST THE AMODAL GROUND TRUTH; ID-OFF IS THE FRACTION OF BURN FRAMES THE ENTERING IDENTITY IS ABSENT. BEST PER COLUMN IN BOLD.

Method	clear↑	Rcv@.5↑	heavy↑	total↑	ID-off↓
PEG (ours)	**93.0**	**76.5**	**74.1**	**77.6**	**18.5**
ByteTrack	88.5	8.6	0.0	0.0	94.2
OC-SORT	90.8	7.8	0.0	0.0	93.6
Deep-OC-SORT	90.8	7.9	0.0	0.0	94.7
DiffMOT	89.9	9.1	0.0	0.0	94.0
BoT-SORT	88.8	9.1	0.0	0.0	94.2
HybridSORT	90.0	7.4	0.0	0.0	93.6
TrackTrack	89.4	5.3	0.0	0.0	96.6
DIP	91.7	8.8	1.4	0.0	97.3

TABLE IV
OPERATING-POINT SWEEP (TOP) AND ADDITIVE ABLATION (BOTTOM) ON THE SYNTHETIC TWINS, FULL-SCENE MOT. THE SIMPLE GATE (h=0.4) IS THE DEPLOYED MODEL.

Configuration	MOTA↑	FP↓	Rcll↑	IDF1↑
gate h=0.2	72.00	2257	85.78	74.71
gate h=0.3	72.70	2124	85.68	75.04
gate h=0.4 (deployed)	73.24	2024	85.61	75.29
gate h=0.5	73.74	1939	85.60	75.50
gate h=0.6	**73.82**	1926	85.60	**75.53**
+ matched-filter confirm	73.42	1994	85.61	75.36
+ CUSUM	67.53	3035	**86.03**	73.10
+ all three	71.56	2353	85.92	74.54
− gate (plain coast, cap 12)	71.69	**1889**	83.23	74.35

real burnt-present positives (221 each). The gate emits all positives; on reflections its high threshold already rejects most (27.6% false emission, as a diffuse glow rarely clips 40% of a box), and the matched filter barely helps (21.3%) while costing recall (100→97.7%, a total burn leaving no *compact* core). The hard case is a foreign headlamp on a departed box: it defeats the threshold outright (100%) and the matched filter only dents it (86.4%), since the core is real but not this worker's, so resolving it needs position, not photometry (Limitations).

VI. CONCLUSION

Tracking through headlamp saturation is, on near-static workers, a detection problem, not a localization one: a constant-velocity coast already predicts the body, and the open question is whether a missed detection is a recoverable burn or a departure. PEG answers it with a single analytic quantity, the saturated fraction of the predicted body box, gating amodal emission so that burnt workers are recovered at the recall of a blind coast but 56% fewer false positives and a higher MOTA than not recovering at all, with no appearance model and one threshold in the decision. The glare-aware front-end includes a learned visibility head, but it is blind during a total burn, so the recovery rests on the analytic gate; classical matched-filter, sequential and innovation refinements do not improve the rule. Limitations: the gate addresses saturation, not foreground occlusion; the burn fraction is a heuristic a *foreign* headlamp on a departed box can fool (Section V), needing position-based gating to resolve; and synthetic burns supply the ground-truth severity.

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